

Aperture Science Test Chamber Construction Guidelines

Purpose

This document provides technical specifications for constructing standard and advanced test chambers within the Enrichment Center.

Chamber Types

Standard Test Chamber

- **Dimensions**: 20m x 20m x 10m
- **Features**: One button, one pressure plate, basic portal surfaces
- **Difficulty**: Introductory (1/5 rating)

Advanced Test Chamber

- **Dimensions**: 30m x 30m x 15m
- **Features**: Multiple buttons, moving platforms, energy pellets
- **Difficulty**: Advanced (3/5 rating)

GLaDOS-Controlled Chamber

- **Dimensions**: Variable (AI-specified)
- **Features**: Adaptive layout, dynamic obstacles, emotional manipulation
- **Difficulty**: Variable (AI adjusts)

Construction Materials

- **Walls**: Aperture Science Portal-Enabled Panels
- **Floors**: Non-slip, high-durability composite
- **Ceilings**: Reinforced steel grating
- **Portal Surfaces**: Moon-dust based coating (Patented)

Step-by-Step Construction

Step 1: Foundation

- Clear area (Radius: 25m minimum)
- Install floor panels (Must be perfectly level)
- Lay electrical conduits
- Connect power grid (Requires 10,000 kW)

Step 2: Wall Installation

- Erect wall frames
- Attach portal-enabled panels
- Seal all seams
- Test surface integrity

Step 3: Automation Setup

1. Install pressure plate sensors
2. Wire button activation mechanisms
3. Set up moving platform paths
4. Calibrate timing sequences

Step 4: AI Integration

1. Connect to GLaDOS network
2. Upload chamber design data
3. Run diagnostic tests
4. Obtain final AI approval

Testing Chamber Verification

Checklist

- ☐ All walls are portal-able
- ☐ Buttons activate correct outputs
- ☐ Time sequences function precisely
- ☐ No escape routes (Unless intended)
- ☐ GLaDOS can observe without interference

Testing Protocol

- Run 50 test scenarios
- Ensure 98% puzzle completion rate
- Verify safety mechanisms
- Adjust difficulty based on test subject performance

Special Considerations

- Keep companion cube access points open
- Ensure weight cubes are available nearby
- Test chamber near emergency exit (Especially for new subjects)
- Monitor for "Chell" arrival (Special handling required)